LAB EXPERIMENT

4. Implement histogram equalization on the given image and compare it with the original image using Open CV.

Code:

```
import cv2
import matplotlib.pyplot as plt

# Read the image
img = cv2.imread(r"C:\Users\Admin\Downloads\images.jpg")

# Check whether the image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Convert image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply histogram equalization
    equalized = cv2.equalizeHist(gray)

    # Display original grayscale image
    plt.subplot(2, 2, 1)
    plt.imshow(gray, cmap="gray")
    plt.title("Original Image")
    plt.axis("off")

    # Display equalized image
    plt.subplot(2, 2, 2)
    plt.imshow(equalized, cmap="gray")
    plt.title("Equalized Image")
    plt.axis("off")

    # Original histogram
    plt.subplot(2, 2, 3)
    plt.hist(gray.ravel(), 256, [0, 256])
    plt.title("Original Histogram")

    # Equalized histogram
    plt.subplot(2, 2, 4)
    plt.hist(equalized.ravel(), 256, [0, 256])
```

```
plt.title("Equalized Histogram")

plt.tight_layout()
plt.show()
```

